

36. Quantization Effects in 5G DSP Hardware

1. Aim

To simulate coefficient quantization in an 8-element digital beamforming processor using 8-bit, 10-bit, and 12-bit word lengths and analyze its effect on beam patterns, side-lobe levels, and quantization errors.

2. Software Used

- MATLAB

3. Objectives

1. To implement beamforming weights.
2. To quantize the beamforming coefficients.
3. To compare the ideal and quantized beam patterns.
4. To measure side-lobe levels.
5. To evaluate quantization errors.
6. To recommend a suitable word length.

4. Procedure

1. Define an 8-element uniform linear antenna array.
2. Set the antenna spacing to $\lambda/2$.
3. Generate complex beamforming weights for a steering angle of 20° .
4. Calculate the ideal beam pattern.
5. Quantize the real and imaginary parts of the coefficients using 8-bit, 10-bit, and 12-bit precision.
6. Calculate the beam pattern for each quantized coefficient set.
7. Plot and compare the ideal and quantized beam patterns.
8. Calculate the side-lobe levels.
9. Calculate the mean-square quantization error.
10. Compare the results for all three word lengths.

5. MATLAB Program

```
clc;  
clear;  
close all;
```

```
% Parameters
```

```
N = 8;
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d = 0.5;
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theta0 = 20;
theta = -90:0.01:90;

% Antenna positions
n = 0:N-1;

% Ideal beamforming weights
w = exp(-1j*2*pi*d*n*sind(theta0)).'/sqrt(N);

% Ideal array factor
AF_ideal = zeros(size(theta));

for k = 1:length(theta)
    a = exp(1j*2*pi*d*n*sind(theta(k))).';
    AF_ideal(k) = abs(w' * a);
end

% Normalize
AF_ideal = AF_ideal / max(AF_ideal);
AF_ideal_dB = 20*log10(AF_ideal + eps);

% Word lengths
bits = [8 10 12];

figure;
plot(theta, AF_ideal_dB, 'LineWidth', 1.5);
hold on;

for b = bits
    scale = 2^(b-1);

    % Quantize real and imaginary parts
    wq_real = round(real(w)*scale)/scale;
    wq_imag = round(imag(w)*scale)/scale;
    wq = wq_real + 1j*wq_imag;

    % Quantization error

```

```

mse = mean(abs(w-wq).^2);

% Quantized array factor
AF = zeros(size(theta));

for k = 1:length(theta)
    a = exp(1j*2*pi*d*n*sind(theta(k))).';
    AF(k) = abs(wq' * a);
end

% Normalize
AF = AF / max(AF);
AF_dB = 20*log10(AF + eps);

% Side-lobe level
peak_angle = theta(find(AF == max(AF),1));
mask = abs(theta - peak_angle) > 15;
SLL = max(AF_dB(mask));

fprintf('%d-bit Quantization:\n', b);
fprintf('MSE = %.10e\n', mse);
fprintf('Side-lobe Level = %.3f dB\n\n', SLL);

plot(theta, AF_dB, 'LineWidth', 1.2);
end

grid on;
xlabel('Angle (degrees)');
ylabel('Normalized Beam Pattern (dB)');
title('Effect of Coefficient Quantization on Beam Pattern');
legend('Ideal', '8-bit', '10-bit', '12-bit');
ylim([-40 5]);

```

6. Output

The MATLAB program produces a beam-pattern graph showing:

- Ideal beam pattern
- 8-bit quantized beam pattern

- 10-bit quantized beam pattern
- 12-bit quantized beam pattern

The command window displays the MSE and side-lobe level for each word length.

7. Result

The coefficient quantization of the 8-element digital beamforming processor was successfully simulated for 8-bit, 10-bit, and 12-bit word lengths. The beam patterns, side-lobe levels, and quantization errors were obtained and compared.

Results Table

Word Length	Quantization Error (MSE)	Side-Lobe Level
8-bit	$\approx 9.26 \times 10^{-6}$	≈ -12.67 dB
10-bit	$\approx 4.66 \times 10^{-7}$	≈ -12.79 dB
12-bit	$\approx 3.68 \times 10^{-8}$	≈ -12.80 dB

